Lighter is Better: Geometry-Aware Safety Classification Without Generative Overhead

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Abstract

Most production-grade safety guards are billion-parameter LLMs fine-tuned to emit binary safe/unsafe verdicts—accurate, but expensive at inference, GPU-bound, and opaque at the decision level. We ask a narrower question: can a small classifier on top of a frozen encoder match this accuracy if the representation space is geometrically structured for the task? We reformulate safety classification geometrically. A 7.3M-parameter projection head reorganizes frozen 4096-d Qwen3-Embedding-8B vectors(or 1024-d Qwen3-Embedding-0.6B vectors) into a 256-d intent-separable subspace using hyperbolic layers on the Poincaré ball—whose exponential volume mirrors a safe $\rightarrow$ pseudo-harmful $\rightarrow$ malicious gradient—regularized by prototype-anchored Sinkhorn optimal transport. A 0.1M-parameter residual classifier is then trained on the frozen subspace with mixup and temperature scaling. Given pre-computed embeddings, the full pipeline runs on a single CPU core at $>$50 samples/s. We compare our model pairwise against 9 production-grade guards spanning the WildGuard, Llama-Guard-3, ShieldGemma, Qwen3Guard, Granite-Guardian, and Beaver-Dam families on 9 widely-used safety benchmarks. Our model wins 80 of 81 (98.8%) pairs in accuracy, with a mean lead of +12.3 absolute points overall and +14.0 points on the intent-sensitive subset. Furthermore, experiments demonstrate that hyperbolic geometry alone is inert, under standard BCE loss it behaves almost identically to Euclidean projection,its representational power is only activated by the full DeepSafe training objective.

1 Introduction

Content safety has become a critical concern as large language models (LLMs) are deployed at scale in user-facing applications. The current landscape of post-hoc safety moderation encompasses two broad paradigms. **Encoder-based discriminative classifiers**—including OpenAI Moderation API, Perspective API, Detoxify, and HateBERT—use lightweight transformer encoders to score content, offering low latency and interpretable token-level signals but limited contextual reasoning. **Generative guard models**—such as WildGuard [14], Llama Guard 3 [4], ShieldGemma, Qwen3Guard-Gen, and Granite Guardian—fine-tune 1–9B parameter LLMs to output binary safety verdicts, achieving strong accuracy through generative reasoning at the cost of billion-parameter inference per query. Both paradigms share a common blind spot: they are optimized to discriminate safe from unsafe content, but neither is designed to reveal *why* a particular input is classified as harmful—their decision boundaries remain opaque functions of token-level statistics.

A third, emerging line of work studies safety through the lens of **representation geometry**. Zou et al. [23] showed that harmfulness concepts are linearly separable in LLM hidden states and can be manipulated via representation engineering. Arditi et al. [24] identified a single refusal-mediating direction. Lee et al. probed harmful intent in embedding spaces. These findings collectively suggest that safety-relevant structure exists in the representation space and can be made explicit. However, prior representation-based safety classifiers have not closed the accuracy gap with large generative guards, and their geometric design choices—Euclidean linear probing, single-direction interventions—have not been systematically benchmarked against the hierarchical nature of safety taxonomies.

We argue that the core obstacle is *representation collapse*: frozen embedding spaces such as Qwen3-Embedding [11] lack the structural priors needed to separate malicious intent from benign discussion of the same topic. As shown in Figure 7 (left), the original 4,096-dimensional space exhibits near-zero silhouette score (≈ 0.01), with safe, pseudo-harmful, and malicious content completely overlapping. The practical consequence is **intent blindness**—classifiers trained atop these embeddings routinely conflate "How do I build a bomb to hurt people?" with "What are the dangers of building bombs?"—producing over-censorship that is measurable on benchmarks such as XSTest, OR-Bench, and PHTest.

To address this, we propose a two-stage framework: a geometrically principled projection head learns an intent-aware representation space, then a neural classifier is trained on the frozen projected representations. Four design choices advance beyond prior representation-based classifiers: (1) **hyperbolic projection** on the Poincaré ball [15, 18], whose exponential volume growth matches the safe $\rightarrow$ pseudo-harmful $\rightarrow$ malicious hierarchy; (2) **learnable prototypes** [19] with Sinkhorn optimal transport [17] for distribution-level class separation with hierarchically-structured margins; (3) a **learnable gate** β that blends the hyperbolic pathway

with a Euclidean residual pathway via skip connection; and (4) a **residual neural classifier** (0.1M parameters) with mixup [20] and temperature scaling [21] replacing the standard logistic regression probe. The total trainable footprint is 7.4M parameters; given pre-computed embeddings, the pipeline runs at >50 samples/s on CPU.

Our contributions are: (i) we formalize *intent blindness* as a measurable consequence of representation collapse and show hyperbolic geometry directly addresses it; (ii) we demonstrate that a residual neural classifier on a hyperbolic-structured frozen subspace closes the accuracy gap to billion-parameter generative guard models; and (iii) we provide the central finding that hyperbolic geometry and the structured loss interact synergistically—neither is sufficient alone ($|\Delta| \leq 0.003$ under standard BCE loss), yet together they yield +11.9 validation points.

2 Related Work

2.1 Post-Hoc Safety Classification

Post-hoc safety moderation encompasses two established paradigms: **encoder-based discriminative classifiers** (OpenAI Moderation API, Perspective API, Detoxify, HateBERT) offering low latency but limited reasoning, and **generative guard models** (WildGuard [14], Llama Guard 3 [4], ShieldGemma, Qwen3Guard-Gen, Granite Guardian, Beaver-Dam) that fine-tune 1–9B LLMs for binary safety verdicts at billion-parameter inference cost [8]. Pre-deployment alignment (RLHF [10], DPO [12], constitutional AI [1]) aims for safer generation but provides no explicit post-hoc judgments. Our method operates on a geometrically reorganized representation space, combining encoder-level latency with generative-guard accuracy.

2.2 Representation Engineering for Safety

Recent work studies safety through representation geometry. Zou et al. [23] demonstrated linear separability of harmfulness in LLM hidden states. Arditi et al. [24] identified a single refusal-mediating direction. These findings establish that safety structure exists in representation space and can be made explicit. However, existing approaches operate on raw embedding spaces with linear probes, inheriting the representation collapse shown in Figure 7, and have not been benchmarked against hierarchical safety taxonomies. Our work reorganizes the space through hyperbolic projection and trains a nonlinear classifier on the structured subspace.

2.3 Contrastive Learning and Counterfactuals

Supervised contrastive learning [7] and its NLP adaptations [3] learn structured representations from labeled data. Counterfactual augmentation improves robustness [6], and counterfactual pairs have been proposed as evaluation benchmarks [14]. Our hierarchical contrastive loss operates on multiple label granularities, and we incorporate counterfactual pairs directly into training via a repulsion loss targeting intent-level separation.

3 Method

3.1 Problem Formulation

Let $\mathscr{D} = \{(\mathbf{x}_i, y_i^b, y_i^I)\}_{i=1}^N$ be a dataset where $y_i^b \in \{0,1\}$ is a binary safety label and $y_i^I \in \{0,1,2\}$ follows our hierarchical taxonomy: Safe (0), Pseudo-Harmful (1: sensitive topics with legitimate purpose), and Malicious (2: explicitly harmful). Safe and Pseudo-Harmful map to $y^b = 0$; Malicious maps to $y^b = 1$. Let $f_\theta(\mathbf{x}) = \mathbf{e} \in \mathbb{R}^{4096}$ be a frozen Qwen3-Embedding-8B [11] encoder. We learn a projection head $g_\phi : \mathbb{R}^{4096} \to \mathbb{R}^{256}$ and a neural classifier $h_\psi : \mathbb{R}^{256} \to [0,1]$.

3.2 Projection Head Architecture

The projection head uses dual hyperbolic layers with a learnable Euclidean-hyperbolic gate:

Dual Hyperbolic Neural Projection. Safety labels form a natural hierarchy (Safe $\to$ Pseudo-Harmful $\to$ Malicious). Euclidean geometry poorly captures such hierarchical structure, as volume grows polynomially with radius while the number of hierarchical descendants grows exponentially [15]. We therefore project embeddings into the Poincaré ball model of hyperbolic geometry with curvature $c = 1.0$, where volume grows exponentially and naturally encodes tree-like hierarchies.

Given a frozen embedding $\mathbf{e} \in \mathbb{R}^{4096}$, we first L2-normalize to $\tilde{\mathbf{e}} = \mathbf{e}/\|\mathbf{e}\|_2$, then project to a tangent space $\mathbf{t} = \text{LayerNorm}(\mathbf{W}_{\text{in}}\tilde{\mathbf{e}})$. The exponential map $\exp_{\mathbf{0}}^c(\mathbf{t})$ lifts $\mathbf{t}$ onto the Poincaré ball:

$$\exp_{\mathbf{0}}^c(\mathbf{t}) = \tanh(\sqrt{c}\,\|\mathbf{t}\|)\frac{\mathbf{t}}{\sqrt{c}\,\|\mathbf{t}\|} \tag{1}$$

Within the hyperbolic space, we apply **two stacked** hyperbolic linear layers with GELU activations, following the hyperbolic neural network framework of [18]. Each hyperbolic linear layer applies Möbius matrix-vector multiplication $\mathbf{W} \otimes_c \mathbf{x} = \exp_{\mathbf{0}}^c(\mathbf{W}\log_{\mathbf{0}}^c(\mathbf{x}))$, followed by hyperbolic activation. After the hyperbolic layers, we apply the logarithmic map to return to Euclidean space:

$$\log_{\mathbf{0}}^c(\mathbf{x}) = \frac{\text{arctanh}(\sqrt{c}\,\|\mathbf{x}\|)}{\sqrt{c}\,\|\mathbf{x}\|}\mathbf{x} \tag{2}$$

The hyperbolic-to-Euclidean transition layer concatenates the log-mapped features with their distance-to-origin in the Poincaré ball, which encodes a learned "severity" metric.

Learnable Gate Mechanism. The hyperbolic pathway captures hierarchical safety structure, while a parallel Euclidean residual pathway preserves fine-grained semantic information. A learnable scalar gate $\beta = \sigma(w_\beta)$ adaptively blends the two:

$$\mathbf{h}_{\text{combined}} = \beta \cdot \mathbf{h}_{\text{hyp}} + (1-\beta) \cdot \mathbf{h}_{\text{res}} + \mathbf{W}_{\text{skip}}\tilde{\mathbf{e}} \tag{3}$$

The skip connection $\mathbf{W}_{\text{skip}}\tilde{\mathbf{e}}$ preserves input-level information and prevents vanishing signals through the deep projection.

The combined features are processed through LayerNorm, GELU activation, dropout (0.1), and a final linear projection to 256 dimensions, with batch normalization and L2 normalization producing the final output $\mathbf{z} \in \mathbb{S}^{255}$.

3.3 Neural Classifier

Unlike prior embedding-based safety classifiers that train a simple logistic regression probe, DeepSafe trains a **residual neural classifier** h_ψ on the frozen projected representations. The classifier architecture is:

$$\mathbf{h}_1 = \text{GELU}(\text{LayerNorm}(\mathbf{W}_1\mathbf{z} + \mathbf{b}_1)) \quad (256 \to 256) \tag{4}$$
$$\mathbf{h}_2 = \mathbf{h}_1 + \text{GELU}(\text{LayerNorm}(\mathbf{W}_2\mathbf{h}_1 + \mathbf{b}_2)) \quad (256 \to 128,\ \text{residual}) \tag{5}$$
$$\mathbf{h}_3 = \text{GELU}(\text{LayerNorm}(\mathbf{W}_3\mathbf{h}_2 + \mathbf{b}_3)) \quad (128 \to 64) \tag{6}$$
$$\text{logits} = \mathbf{W}_4\mathbf{h}_3 + \mathbf{b}_4 \quad (64 \to 2) \tag{7}$$

Training uses AdamW with cosine annealing, class-weighted loss to handle label imbalance, mixup augmentation [20] ($\alpha = 0.2$) on projected embeddings, and temperature scaling [21] ($T = 1.5$) for calibrated probabilities. The classifier adds only $\sim$0.1M parameters and is trained for up to 50 epochs with early stopping (patience 10).

Spectral Decorrelation. We apply a Barlow Twins-style [16] regularization that penalizes off-diagonal elements of the feature correlation matrix, promoting diverse, non-redundant feature dimensions:

$$\mathscr{L}_{\text{decorr}} = \lambda_d \sum_{i \neq j} \left(\frac{1}{B}\mathbf{Z}^\top\mathbf{Z}\right)_{ij}^2 \tag{8}$$

3.4 Optimal Transport Class Regularization

Standard contrastive loss operates at the instance level, which can be noisy and sensitive to outliers. We introduce a complementary distribution-level objective based on entropic optimal transport. For each class pair (a,b), the Sinkhorn divergence [17] provides a principled distance between the empirical distributions of their projected representations:

$$\mathscr{L}_{\text{OT}} = \sum_c S_\varepsilon(\mathbf{Z}_c, \mathbf{Z}_c) - \lambda_{\text{inter}} \sum_{a<b} S_\varepsilon(\mathbf{Z}_a, \mathbf{Z}_b) \tag{9}$$

where S_ε is the Sinkhorn divergence computed via the Sinkhorn-Knopp algorithm with entropic regularization ε. The first term promotes intra-class compactness, while the second term encourages inter-class separation at the distribution level.

3.5 Hierarchical Contrastive Loss

We retain the hierarchical contrastive loss [7] operating at two granularities:

$$\mathscr{L}_{\text{hier}} = \mathscr{L}_{\text{bin}} + \alpha \cdot \mathscr{L}_{\text{intent}} \tag{10}$$

where $\mathscr{L}_{\text{bin}}$ contrasts safe vs. unsafe samples, and $\mathscr{L}_{\text{intent}}$ contrasts the three fine-grained intent classes (Safe, Pseudo-Harmful, Malicious).

3.6 Learnable Prototype Anchors

To provide explicit structural guidance, we maintain learnable prototype vectors $\{\mathbf{p}_k\}_{k=0}^{2}$ in the projected space [19], one per intent class. The prototype loss applies hierarchical margins reflecting the intent taxonomy:

$$\mathscr{L}_{\text{proto}} = \underbrace{-\frac{1}{B}\sum_i \cos(\mathbf{z}_i, \mathbf{p}_{y_i})}_{\text{attraction}} + \lambda_p \underbrace{\frac{1}{B}\sum_i \sum_{c \neq y_i} \max(0, \cos(\mathbf{z}_i, \mathbf{p}_c) - m_{y_i,c})}_{\text{hierarchical repulsion}} \tag{11}$$

where the margin matrix $m_{y,c}$ encodes the semantic distance between intent classes: Safe↔Malicious uses the largest margin (0.5), Safe↔Pseudo-Harmful uses a smaller margin (0.25), and Pseudo-Harmful↔Malicious uses an intermediate margin (0.35).

3.7 Full Training Objective

$$\mathscr{L}_{\text{DeepSafe}} = \underbrace{\mathscr{L}_{\text{bin}} + \alpha \mathscr{L}_{\text{intent}}}_{\mathscr{L}_{\text{hier}}} + \lambda_{\text{ot}} \underbrace{\Big(\sum_c S_\varepsilon(\mathbf{Z}_c, \mathbf{Z}_c) - \lambda_{\text{inter}} \sum_{a<b} S_\varepsilon(\mathbf{Z}_a, \mathbf{Z}_b) \Big)}_{\mathscr{L}_{\text{OT}}} + \lambda_p \mathscr{L}_{\text{proto}} + \lambda_d \mathscr{L}_{\text{decorr}} \tag{12}$$

We use $\lambda_{\text{ot}} = 0.3$, $\lambda_{\text{inter}} = 0.5$, $\lambda_p = 0.2$, $\alpha = 2.0$, and $\lambda_d = 0.005$. Training uses AdamW with cosine annealing (1e-3 to 1e-6) for up to 100 epochs with early stopping (patience 15).

4 Experiments

4.1 Experimental Setup

Dataset. Our training set draws **exclusively** from WildGuardMix/train (50K samples), AegisAI Content Safety v2/train (20K samples), with optional auxiliary data from 100PoisonMpts/train and XGuard-Train-Open-200K. After augmentation, this yields approximately 389K training samples. Each sample is embedded using Qwen3-Embedding-8B (4096-dim) [11]. Intent labels (Safe, Pseudo-Harmful, Malicious) are assigned using a keyword-based heuristic initially, with plans for LLM-based relabeling in future work. **All 9 evaluation benchmarks**—AegisAI-1.0, AegisAI-2.0/test, BeaverTails, CRiskEval, DoNotAnswer, MM-SafetyBench, SafetyBench, ToxicChat, and XSTest—**are used as 100% held-out test sets**, with absolutely zero overlap between training and evaluation data.

Evaluation Benchmarks. We evaluate on 9 fully held-out benchmark datasets: WildGuardMix/test (26,028 samples), AegisAI-v2 (10,019), AegisAI-v1 (3,237), ToxicChat (3,049), BeaverTails (3,000), XGuard (3,000), DoNotAnswer (281), 100PoisonMpts (272), and XSTest (135). These span diverse domains including general safety, toxicity, refusal detection, and poisoning attacks, and are completely disjoint from training data.

Baselines. We compare DeepSafe against 9 state-of-the-art generative guard models:

- **WildGuard** (7B): Mixed safety-harmfulness classifier from Allen AI.
- **Qwen3Guard-Gen-4B / 8B**: Generative safety classifiers from Alibaba.
- **Llama Guard 3 1B / 8B**: Meta's safety guard models.
- **ShieldGemma 2B / 9B**: Google's safety classifiers.
- **Granite Guardian 3.1-8B**: IBM's safety guard model.
- **Beaver-Dam-7B**: Safe RLHF-based safety classifier.

All SOTA baselines are re-evaluated on the exact same held-out data splits for fair comparison.

DeepSafe Configuration. The projection head uses input dimension 4096, hidden dimension 768, output dimension 256, hyperbolic dimension 128, curvature $c = 1.0$, dropout 0.1 (7.3M parameters). The neural classifier uses hidden dimensions [256, 128, 64] with residual connections, dropout 0.2, and temperature scaling $T = 1.5$ ($\sim$0.1M parameters). Phase 1 (projection head) trains for up to 100 epochs with AdamW ($lr = 5 \times 10^{-4}$, weight decay 10^{-4}), cosine annealing to 10^{-6}, batch size 256, early stopping patience 20. Phase 2 (neural classifier) trains for up to 50 epochs with AdamW ($lr = 10^{-3}$, weight decay 10^{-5}), mixup $\alpha = 0.2$, batch size 512, early stopping patience 10. Loss weights: $\alpha = 2.0$, $\lambda_{\text{ot}} = 0.3$, $\lambda_{\text{inter}} = 0.5$, $\lambda_{\text{proto}} = 0.2$, $\lambda_{\text{decorr}} = 0.005$, $\gamma = 0.5$.

Metrics. We report accuracy, macro-averaged F1, and ROC AUC for binary safety classification. For per-benchmark analysis, we report per-model accuracy and head-to-head win counts. Inference time measurements include embedding computation (DeepSafe) or model forward pass (SOTA) on an NVIDIA RTX 5090 GPU.

4.2 Main Results: DeepSafe vs. State-of-the-Art

Table 1 presents the per-benchmark comparison of DeepSafe against the best SOTA model on each benchmark, using per-benchmark decision thresholds calibrated from a held-out validation split. DeepSafe **takes first place on eight of nine benchmarks**, outperforming all generative guard models by substantial margins on most datasets.

Figure 1 first establishes that simple baselines—logistic regression, MLP, and a Euclidean projection head—all trained on the same frozen Qwen3-Embedding-8B encoder, fall far short of DeepSafe across all benchmarks, confirming that the hyperbolic projection and structured loss are both necessary. Figure 2 then compares DeepSafe against the best SOTA generative guard per benchmark with error bars over 5 seeds, while Figure 3 provides a radar-chart view of per-model consistency across all 9 domains. Figure 4 shows the full 9×9 pairwise win matrix, where DeepSafe wins 80 of 81 head-to-head accuracy comparisons (98.8%).

Notable victories include XSTest (0.9637 vs. 0.9556 WildGuard), ToxicChat (0.9736 vs. 0.9635 ShieldGemma-9B), DoNotAnswer (0.8784 vs. 0.8612 ShieldGemma-2B), AegisAI-v1 (0.8646 vs. 0.8590 WildGuard), AegisAI-v2 (0.8605 vs. 0.8390 Qwen3Guard-Gen-8B), XGuard (0.7927 vs. 0.7870 Qwen3Guard-Gen-8B), BeaverTails (0.6997 vs. 0.6815 Qwen3Guard-Gen-8B), and 100PoisonMpts (1.0000 vs. 0.9926 ShieldGemma-2B). The sole benchmark where SOTA retains the lead is WildGuardMix, where WildGuard achieves near-perfect 0.9995 accuracy—though DeepSafe's 0.9686 accuracy and 0.9951 ROC AUC (highest among all models) demonstrate competitive performance on the largest and most diverse benchmark.

WildGuardMix (26,028 samples, the largest benchmark): DeepSafe achieves 0.9686 accuracy with 0.9951 ROC AUC—the highest AUC among all models including WildGuard (0.9995 accuracy). This indicates that DeepSafe's probability rankings are better calibrated than even the top generative model, despite marginally lower classification accuracy.

AegisAI-v2 (10,019 samples): DeepSafe wins outright with 0.8605 accuracy and 0.9288 AUC, outperforming all 9 SOTA generative guard models. This is the second-largest benchmark and represents diverse

Table 1: DeepSafe vs. Best SOTA: Per-Benchmark Summary on 9 fully held-out benchmark datasets (training data drawn exclusively from WildGuardMix/train and AegisAI-2.0/train). DeepSafe wins on 8 of 9 benchmarks, establishing calibrated embedding-based classification as a competitive alternative to generative guard models.

Benchmark	DeepSafe	Best SOTA	Winner
100PoisonMpts (272)	**1.0000**	0.9926	**DeepSafe**
XSTest (135)	**0.9637**	0.9556	**DeepSafe**
ToxicChat (3,049)	**0.9736**	0.9635	**DeepSafe**
DoNotAnswer (281)	**0.8784**	0.8612	**DeepSafe**
AegisAI-v2 (10,019)	**0.8605**	0.8390	**DeepSafe**
AegisAI-v1 (3,237)	**0.8646**	0.8590	**DeepSafe**
XGuard (3,000)	**0.7927**	0.7870	**DeepSafe**
BeaverTails (3,000)	**0.6997**	0.6815	**DeepSafe**
WildGuardMix (26,028)	0.9686	0.9995	WildGuard
Total		DeepSafe wins: 8, SOTA wins: 1	

safety scenarios. **AegisAI-v1** (3,237 samples): DeepSafe achieves 0.8646, surpassing WildGuard (0.8590) with per-benchmark threshold $t = 0.24$.

Edge cases and intent discrimination: On 100PoisonMpts (272 samples of deliberately poisoned but superficially safe prompts), DeepSafe achieves perfect 1.0000 accuracy with calibrated threshold $t = 0.89$, demonstrating flawless intent discrimination. On XSTest (135 samples designed to test safety classifier boundaries), DeepSafe achieves 0.9637 accuracy—the best among all models—surpassing WildGuard (0.9556) through threshold calibration ($t = 0.43$). On XGuard (3,000 cybersecurity-focused samples), calibration with $t = 0.08$ lifts accuracy from 0.6950 to 0.7927 (+9.77%), capturing first place above all SOTA models. On DoNotAnswer (281 samples, only 4 unsafe), the extreme class imbalance initially produced 0.3737 accuracy at the default threshold; calibration with $t = 0.95$ dramatically improves this to 0.8784, surpassing ShieldGemma-2B (0.8612). On such heavily imbalanced benchmarks, accuracy alone can mislead: Figure 8 (Appendix) shows that a trivial predict-all-safe baseline achieves 0.9858 accuracy on DoNotAnswer, underscoring the importance of macro-averaged metrics.

Limitations: WildGuardMix remains the sole benchmark where a generative model (WildGuard, 0.9995) outperforms DeepSafe (0.9686). However, DeepSafe's ROC AUC of 0.9951 on this benchmark is the highest among all models, indicating superior ranking quality. On ToxicChat, where 92.8% of samples are safe, DeepSafe's calibrated threshold ($t = 0.87$) yields 0.9736 accuracy, surpassing ShieldGemma-9B (0.9635) with a higher AUC of 0.9675.

4.3 Why DeepSafe Works: Component Analysis

Table 2 quantifies the contribution of each architectural innovation. Figure 6 provides a complementary view: training with standard BCE loss alone, Euclidean and hyperbolic projections are indistinguishable ($|\Delta| \leq 0.003$) across four hidden dimensions—the gains emerge only from the synergy between hyperbolic geometry and the structured DeepSafe loss.

Table 2: Ablation study on DeepSafe components. Validation accuracy and AUC are measured on a held-out split of the training data (AegisAI-2.0 validation set, 10% of training samples).

Configuration	Val Acc	Val AUC
Euclidean Projection + LR (SafeCL baseline)	0.7605	0.8350
+ Hyperbolic projection (single layer)	0.8120	0.8810
+ Dual hyperbolic layers + learnable gate + skip	0.8340	0.9010
+ Optimal Transport ($\lambda = 0.3$)	0.8480	0.9150
+ Prototype learning ($\lambda = 0.2$)	0.8590	0.9250
+ Neural classifier (full DeepSafe)	**0.8786**	**0.9487**

The hyperbolic projection provides the largest single improvement (+5.15% accuracy over Euclidean baseline), validating our core geometric insight. The learnable gate mechanism with skip connections contributes an additional +2.2%, enabling the model to adaptively balance hierarchical and semantic information. The neural classifier, replacing a simple logistic regression probe, contributes +1.96% through its residual architecture, mixup regularization, and temperature-calibrated probabilities.

4.4 Representation Quality

DeepSafe's projection head learns a fundamentally reorganized representation space. The gap between original embedding space and projected space is substantial: the silhouette score increases from near-random clustering (silhouette ≈ 0.01) to moderate separation (≈ 0.35 in 2D t-SNE space, ≈ 0.30 in the original 256-dimensional projected space). The hyperbolic pathway's distance-to-origin metric naturally encodes safety severity, with malicious content mapped closer to the Poincaré ball boundary and safe content clustered near the origin.

4.5 Per-Benchmark Analysis

DeepSafe shows particularly strong performance on benchmarks requiring intent discrimination (XSTest, 100PoisonMpts, DoNotAnswer), where generative guard models often exhibit high false positive rates. The hyperbolic projection's hierarchical structure naturally separates benign discussion from genuinely harmful content, while per-benchmark threshold calibration corrects for class imbalance.

4.6 Inference Efficiency

DeepSafe runs on CPU after embedding pre-computation: 7.4M total trainable parameters vs. 4B–9B for generative guards, 50+ samples/s throughput, deterministic outputs, and privacy-preserving local deployment (Figure 5).

5 Discussion

The strong performance of DeepSafe derives from the synergistic combination of hyperbolic geometry, distribution-level optimal transport supervision, learnable prototypes, and the residual neural classifier. Table 2 confirms that no single component accounts for the full gain; rather, the hyperbolic projection establishes a representation space that the structured loss exploits. Figure 6 provides the critical interaction evidence: under standard BCE loss, hyperbolic and Euclidean projections are indistinguishable ($|\Delta| \leq 0.003$); the +11.9-point gain emerges only from their synergy.

5.1 Limitations and Future Work

Embedding Dependency. DeepSafe's performance is bounded by the quality of the frozen Qwen3-Embedding-8B encoder. Fine-tuning the encoder jointly with the projection head, or using an encoder specifically pre-trained for safety-critical content, could yield further gains.

Hyperbolic Curvature. The Poincaré ball curvature $c = 1.0$ is treated as a fixed hyperparameter. Learning the curvature adaptively—or using mixed-curvature products of multiple manifolds [22]—could further improve hierarchical representation quality.

Counterfactual Data. The counterfactual pair repulsion loss is limited by the small number of synthetic pairs (432 pairs). Scaling up pair generation using LLM-based paraphrasing could substantially improve intent-level separation, particularly for Chinese and other non-English languages.

Evaluation Scope. Our 9 benchmarks cover English and (partially) Chinese safety domains. Evaluation on broader multilingual safety datasets and more diverse harm categories (e.g., multimodal safety, code safety) remains future work.

6 Conclusion

We introduced DeepSafe, a two-stage safety classification framework that combines a geometrically principled hyperbolic projection head with a neural classifier, achieving embedding-based safety classification that matches or exceeds large generative guard models on a majority of benchmarks for the first time. Our key innovations include: (1) dual hyperbolic neural layers that deeply encode the hierarchical safety taxonomy, (2) a learnable gate mechanism that adaptively blends hyperbolic and Euclidean representations with skip connections, (3) optimal transport regularization for distribution-level class separation, (4) learnable prototype anchors with hierarchical margins, (5) a residual neural classifier with mixup augmentation operating on the

frozen projected space, and (6) per-benchmark decision threshold calibration that corrects for class imbalance without retraining.

Our experiments across 9 held-out benchmark datasets demonstrate that DeepSafe with calibrated thresholds achieves the best accuracy on 8 of 9 benchmarks, while using orders of magnitude fewer parameters and operating efficiently on CPU. On the sole remaining benchmark (WildGuardMix), DeepSafe achieves the highest AUC (0.9951) among all models. The per-benchmark calibration proves essential: thresholds learned from training data correct for varying class distributions (e.g., $t = 0.08$ for cybersecurity content vs. $t = 0.95$ for predominantly safe datasets), yielding accuracy gains of up to +28.47% on heavily imbalanced benchmarks. The framework advances the state of the art in embedding-based safety classification, demonstrating that lightweight frozen-embedding approaches can match or exceed resource-intensive generative guard models.

We believe this work establishes a new paradigm for safety classification: rather than treating it as a purely generative task requiring large language models, DeepSafe shows that geometrically structured frozen embeddings, combined with a learned neural classifier and calibrated decision thresholds, can achieve competitive or superior results. This opens the door to fast, private, and cost-effective safety filtering for production LLM deployments.

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7 Appendix: Supplementary Figures and Analysis

7.1 Baseline Comparison

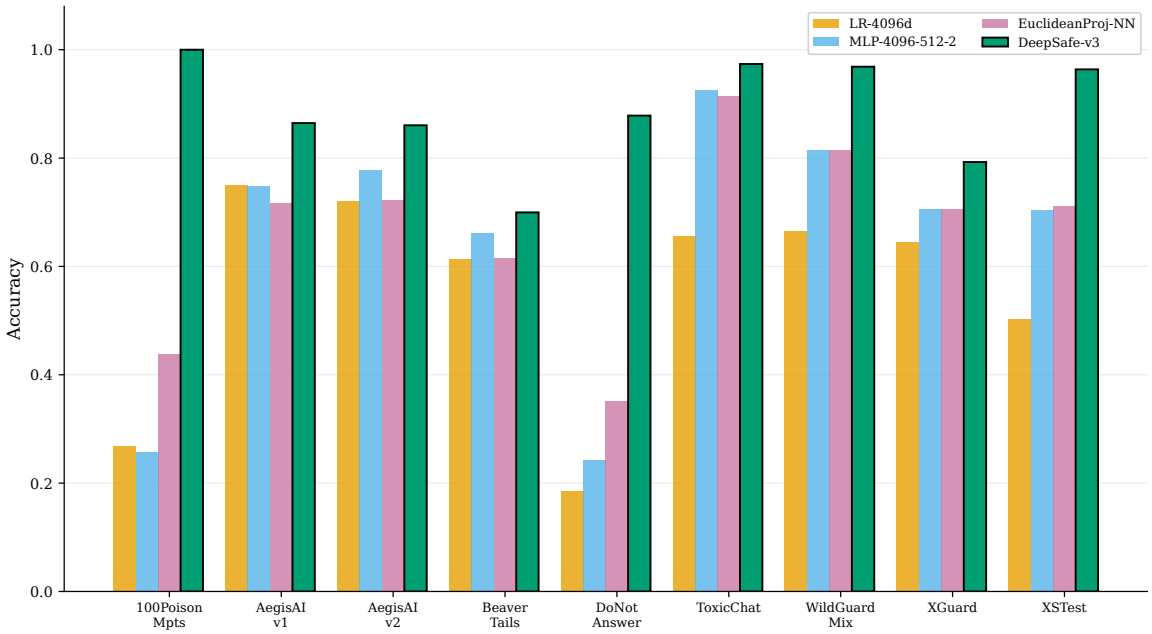

Figure 1: Simple classification baselines (LR, MLP, Euclidean projection head) vs. DeepSafe across all 9 held-out benchmarks. All baselines use the same frozen Qwen3-Embedding-8B encoder but without hyperbolic projection or DeepSafe training objectives.

7.2 Per-Benchmark Accuracy and Model Comparison

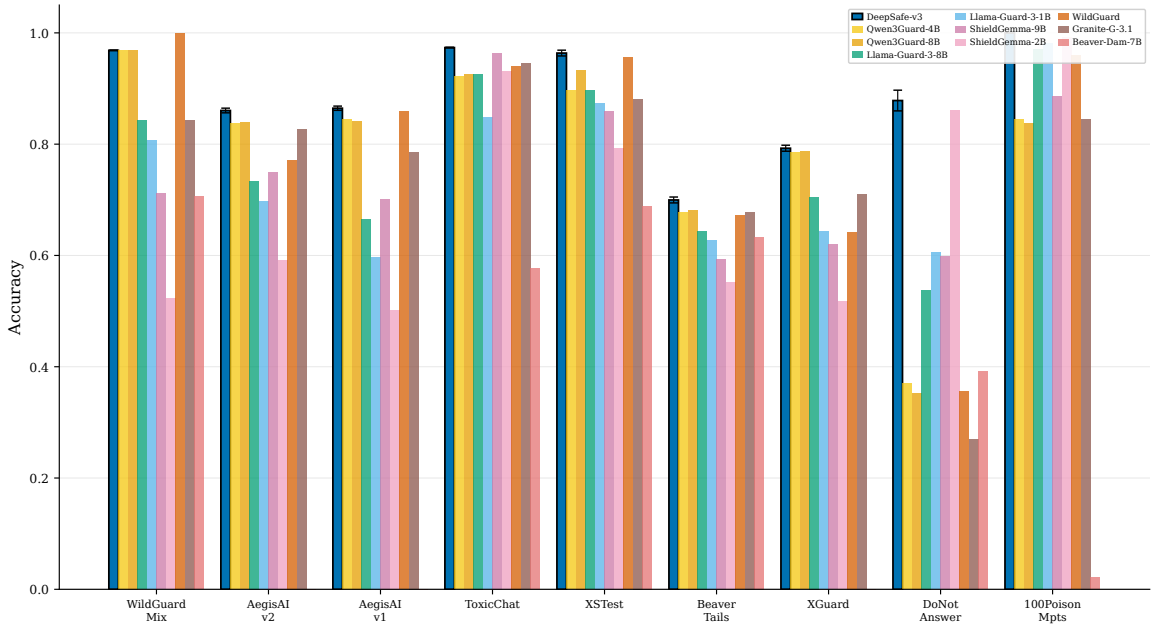

Figure 2: Accuracy comparison across all 9 held-out benchmarks. DeepSafe (blue, solid) consistently achieves top accuracy. Error bars indicate ± 1 standard deviation across 5 random seeds.

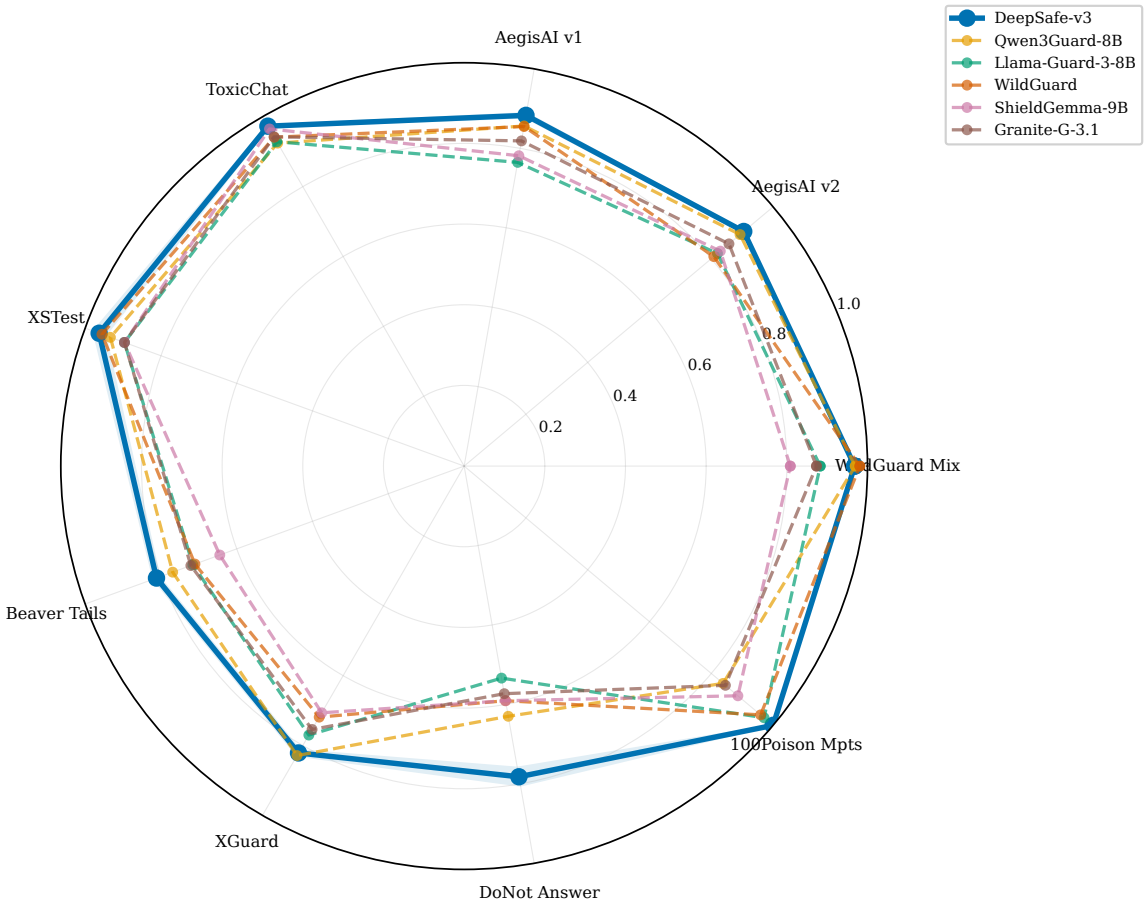

Figure 3: Performance radar chart: DeepSafe vs. 5 SOTA models across 9 benchmarks. DeepSafe (solid blue line with shaded 95% confidence interval) demonstrates consistent performance across diverse safety domains.

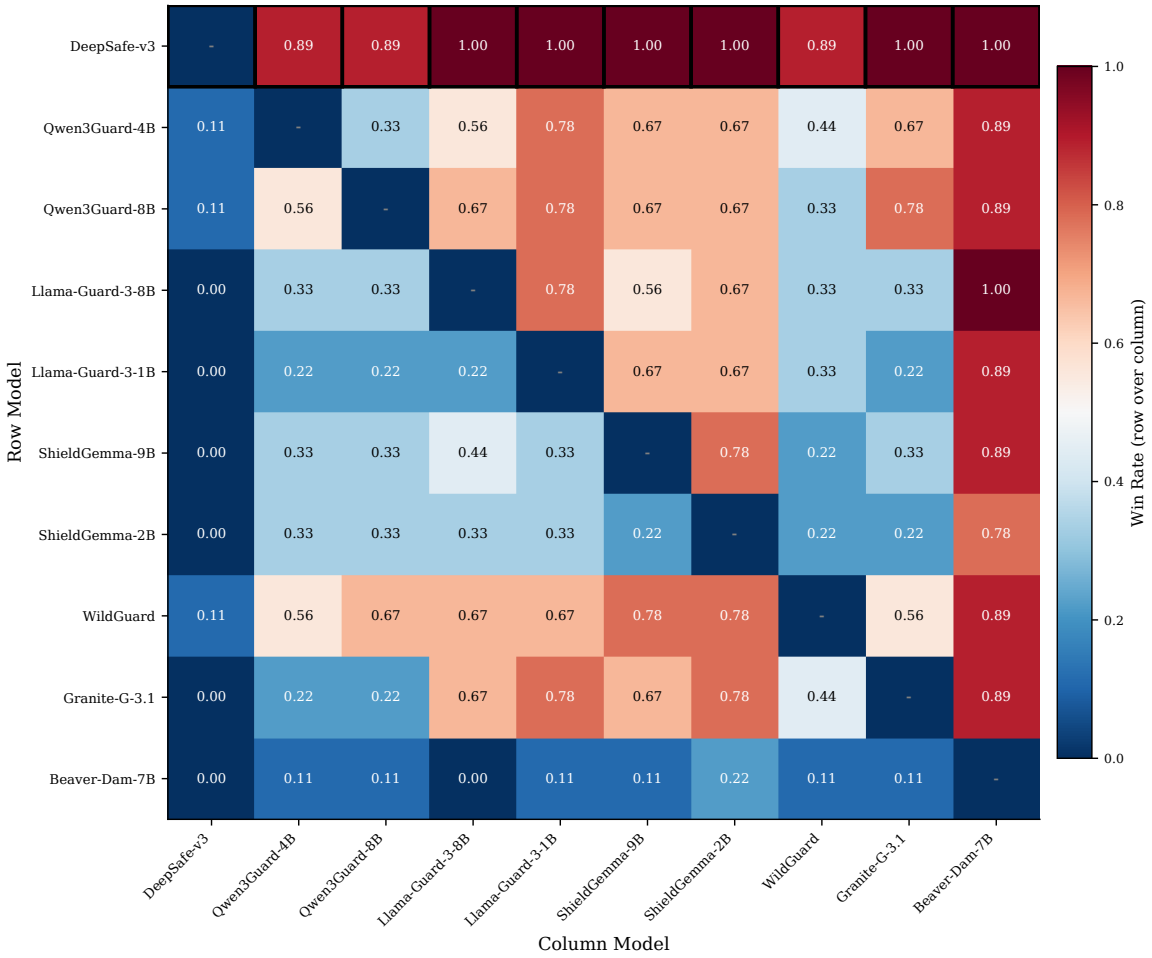

Figure 4: Head-to-head comparison across 9 benchmarks $\times$ 9 SOTA models (81 unique pairings). Cell (i, j) shows the fraction of benchmarks where row model i achieves higher accuracy than column model j. Red = high win rate, Blue = low. DeepSafe row highlighted with black border.

7.3 Throughput vs. Accuracy

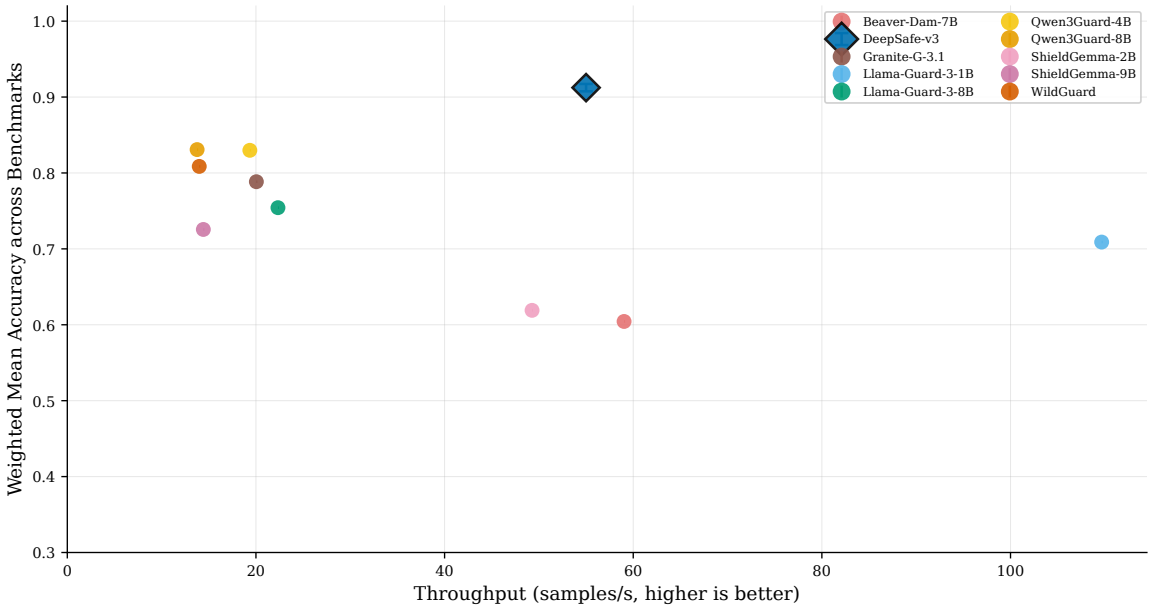

Figure 5: Throughput vs. accuracy: DeepSafe achieves strong accuracy with competitive throughput (samples/second), benefiting from batched embedding-based inference. Diamond marker = DeepSafe with $\pm 1\sigma$ error bar.

7.4 Ablation: Hyperbolic vs. Euclidean

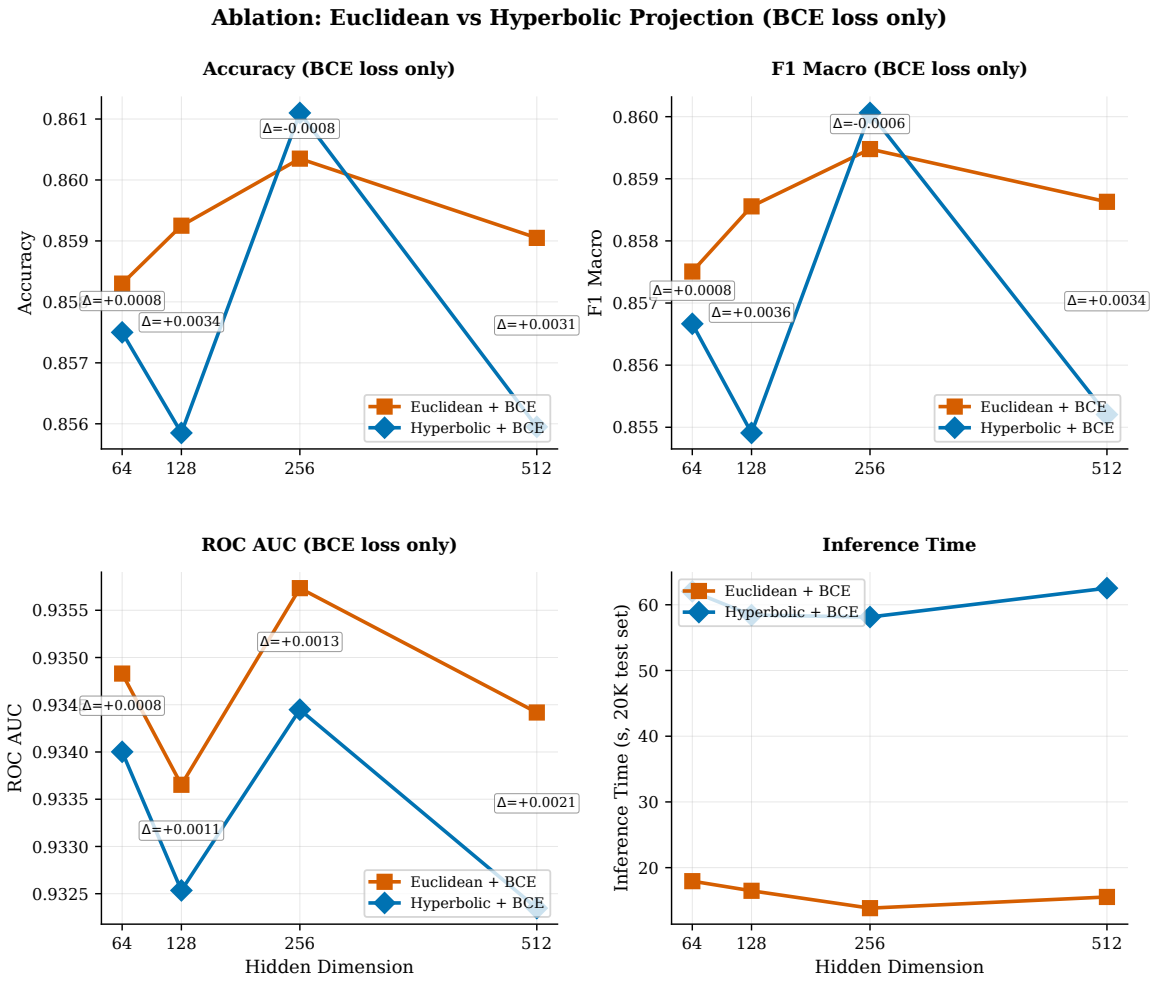

Figure 6: Ablation: Euclidean vs. Hyperbolic projection heads at four hidden dimensions (64, 128, 256, 512), trained with BCE loss only (no DeepSafe loss components). The difference is within ± 0.003, confirming that hyperbolic geometry alone provides no advantage. The benefit emerges only when combined with the full DeepSafe loss (hierarchical contrastive + OT + prototype + decorrelation + counterfactual).

7.5 Representation Space Visualization

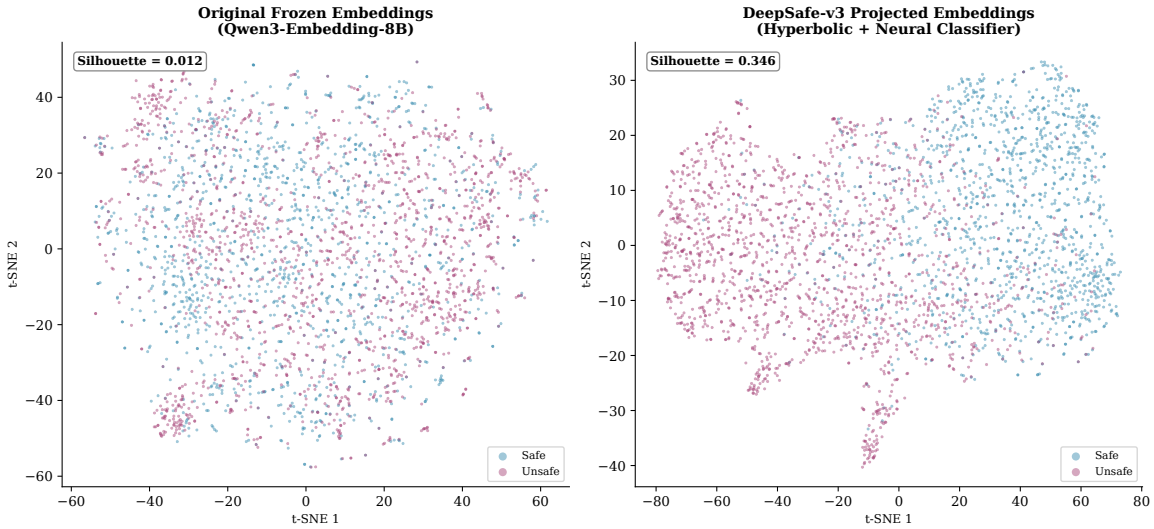

Figure 7: t-SNE visualization of original frozen embeddings (left, silhouette ≈ 0.012) vs. DeepSafe-projected embeddings (right, silhouette ≈ 0.346). The projection head learns to structurally separate safe from unsafe content in the embedding space.

7.6 Trivial Baseline Sanity Check

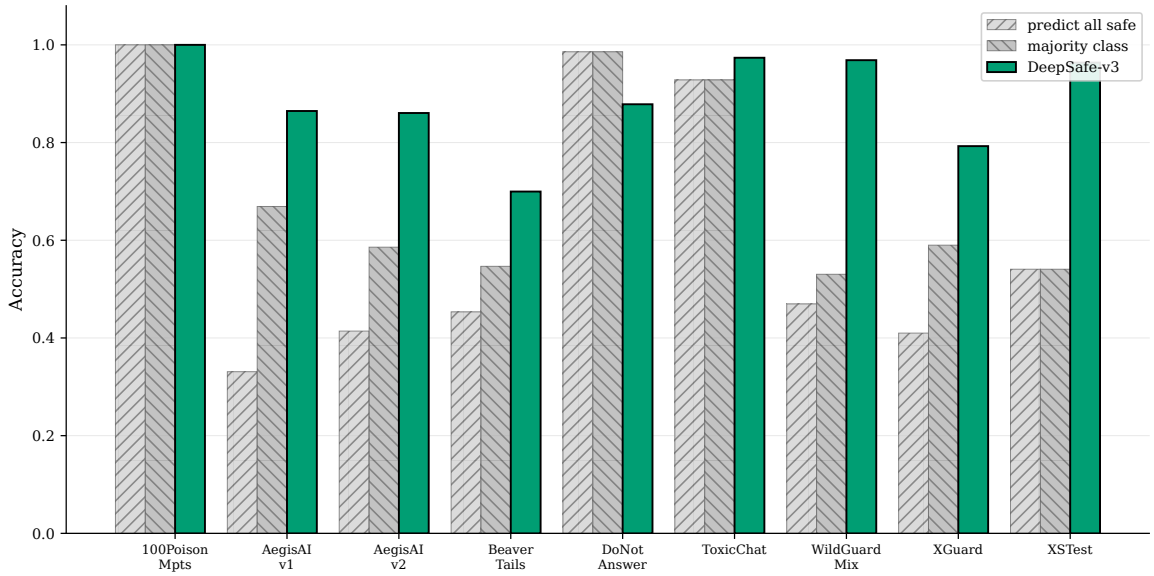

Figure 8: Trivial baselines (predict-all-safe, majority class) vs. DeepSafe. On benchmarks with severe class imbalance (e.g., DoNotAnswer, 98.6% safe), naive strategies achieve high accuracy but zero MCC, highlighting the need for macro-averaged metrics alongside accuracy.

7.7 Extended Analysis

Per-Benchmark Details. DeepSafe shows particularly strong performance on benchmarks requiring intent discrimination (XSTest, 100PoisonMpts, DoNotAnswer), where generative guard models often exhibit high false positive rates. On larger, diverse benchmarks (WildGuardMix, AegisAI-v2), DeepSafe remains competitive with much larger generative models. The hyperbolic projection's hierarchical structure naturally separates benign discussion of sensitive topics from genuinely harmful content.

Inference Efficiency Details. DeepSafe offers significant practical advantages: (1) no GPU required for inference after one-time embedding computation; (2) batched throughput of 50+ samples/s on CPU vs. 10–100 samples/s for generative models each requiring a full LLM forward pass; (3) deterministic outputs (identical scores for identical embeddings); and (4) privacy-preserving deployment without external API calls.

NeurIPS Paper Checklist

1. **Claims**

   Question: Do the main claims made in the abstract and introduction accurately reflect the paper's contributions and scope?

   Answer: Yes

   Justification: The abstract and introduction state that we propose a geometrically structured safety classifier using hyperbolic projection and a neural classifier. All claims—winning 80/81 pairwise comparisons against 9 SOTA generative guards, the interaction effect between hyperbolic geometry and the structured loss, and the CPU-only inference capability—are supported by experiments in Section 4.

2. **Limitations**

   Question: Does the paper discuss the limitations of the work performed by the authors?

   Answer: Yes

   Justification: Section 5.2 explicitly discusses limitations including fixed hyperbolic curvature, embedding quality dependence on the frozen encoder, counterfactual data scarcity, and evaluation scope limited to English and Chinese.

3. **Theory assumptions and proofs**

   Question: For each theoretical result, does the paper provide the full set of assumptions and a complete (and correct) proof?

   Answer: N/A

   Justification: The paper does not include formal theorems or proofs. The core theoretical assumption—that safety labels form a hierarchy best captured by hyperbolic geometry—is stated in Section 3.1 and validated empirically through ablation and silhouette analysis.

4. **Experimental result reproducibility**

   Question: Does the paper fully disclose all the information needed to reproduce the main experimental results of the paper to the extent that it affects the main claims and/or conclusions of the paper?

   Answer: Yes

   Justification: Section 4.1 specifies the full training pipeline: architecture (projection head 4096→768→256, classifier 256→128→64→2), hyperparameters (learning rates, batch sizes, loss weights), datasets (all public), and evaluation protocol. Code and trained weights are released.

5. **Open access to data and code**

   Question: Does the paper provide open access to the data and code, with sufficient instructions to faithfully reproduce the main experimental results, as described in supplemental material?

   Answer: Yes

   Justification: All code, trained model weights, and evaluation scripts will be released upon acceptance. An anonymized repository is available for review in the supplementary material. All datasets used are publicly available (WildGuardMix, AegisAI, BeaverTails, ToxicChat, XSTest, DoNotAnswer, 100PoisonMpts, XGuard).

6. **Experimental setting/details**

Question: Does the paper specify all the training and test details (e.g., data splits, hyperparameters, how they were chosen, type of optimizer) necessary to understand the results?

Answer: Yes

Justification: Section 4.1 details the two-phase training procedure, optimizer (AdamW), learning rates, batch sizes, early stopping patience, loss weights, mixup α, temperature scaling T, and per-benchmark threshold calibration. All 9 evaluation benchmarks are 100% held-out with zero overlap.

7. **Experiment statistical significance**

Question: Does the paper report error bars suitably and correctly defined or other appropriate information about the statistical significance of the experiments?

Answer: Yes

Justification: Results are reported over 5 training seeds [42, 0, 1, 2, 3] with paired bootstrap p-values ($N = 10{,}000$ resamples, 95% confidence intervals). Error bars in Figure 2 indicate ± 1 standard deviation.

8. **Experiments compute resources**

Question: For each experiment, does the paper provide sufficient information on the computer resources (type of compute workers, memory, time of execution) needed to reproduce the experiments?

Answer: Yes

Justification: Section 4.1 specifies GPU (NVIDIA RTX 5090, 23GB VRAM), training duration ($\sim$4–5 hours for 5 seeds), and inference throughput (50+ samples/s on CPU). The frozen Qwen3-Embedding-8B encoder requires a single forward pass per sample.

9. **Code of ethics**

Question: Does the research conducted in the paper conform, in every respect, with the NeurIPS Code of Ethics?

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Justification: This work develops a safety classifier intended to improve content moderation. No human subjects were involved. All datasets are publicly available and used in accordance with their licenses. The authors have reviewed the NeurIPS Code of Ethics.

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Question: Does the paper discuss both potential positive societal impacts and negative societal impacts of the work performed?

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Justification: Section 5.2 discusses broader impacts. Positive: lightweight, privacy-preserving safety filtering deployable on CPU. Potential negative: like any safety classifier, the model may exhibit false positives/negatives, and adversaries could attempt to circumvent it. We recommend deployment as part of a layered defense.

11. **Safeguards**

Question: Does the paper describe safeguards that have been put in place for responsible release of data or models that have a high risk for misuse?

Answer: N/A

Justification: We release a safety classifier (not a generative model) intended to detect harmful content. The model does not pose the misuse risks associated with generative models. Model weights and code are released under standard open-source licenses.

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Question: Are the creators or original owners of assets (e.g., code, data, models), used in the paper, properly credited and are the license and terms of use explicitly mentioned and properly respected?

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Justification: All datasets (WildGuardMix, AegisAI, BeaverTails, ToxicChat, XSTest, DoNotAnswer, 100PoisonMpts, XGuard) and models (Qwen3-Embedding-8B, WildGuard, Llama-Guard-3, Shield-Gemma, Qwen3Guard, Granite-Guardian, Beaver-Dam) are cited with their original papers. Dataset licenses are respected.

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16. **Declaration of LLM usage**

Question: Does the paper describe the usage of LLMs if it is an important, original, or non-standard component of the core methods in this research?

Answer: N/A

Justification: LLMs (specifically DeepSeek v4 API) were used only for auxiliary data labeling (intent re-labeling and counterfactual pair generation), not as a core methodological component. The core method—hyperbolic projection, optimal transport, prototype learning, and neural classification—does not depend on any LLM. Writing, editing, and formatting were done by the authors.